

Gym Management System

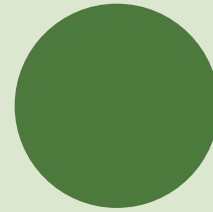
Database Design & Analytics Project

Data Analyst Specialist | NTI

Meet The Team



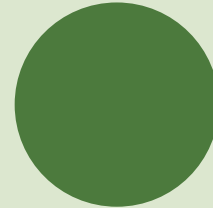
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Abdelrahman Hossam



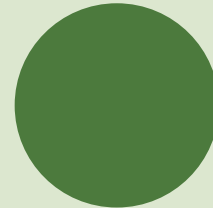
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Agenda

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Introduction

Project Objective

Analyze the Gym Management System data to:

- Monitor member activity and workout behavior
- Evaluate performance across gym branches
- Understand how subscription plans and gym type relate to member engagement

Why?

Understanding member behavior and branch performance is key to improving retention, optimizing operations, and supporting data-driven decisions for the gym business.

Dataset Summary

Source Files (CSV)

Member

users_data.csv

Gym_Branch

gym_locations_data.csv

Facility

derived from gym_locations_data.csv

Gym_Facility

junction table (derived)

Subscription_Plan

subscription_plans.csv

Workout_Session

checkin_checkout_history_updated.csv

5,000

Total Members

~297K

Total Sessions

300,000

Original Rows
(Workout_Session)

Used Tools

Cleaning

MySQL / SQL

Excel & Power Query

Python

Modeling

Excel Power Pivot

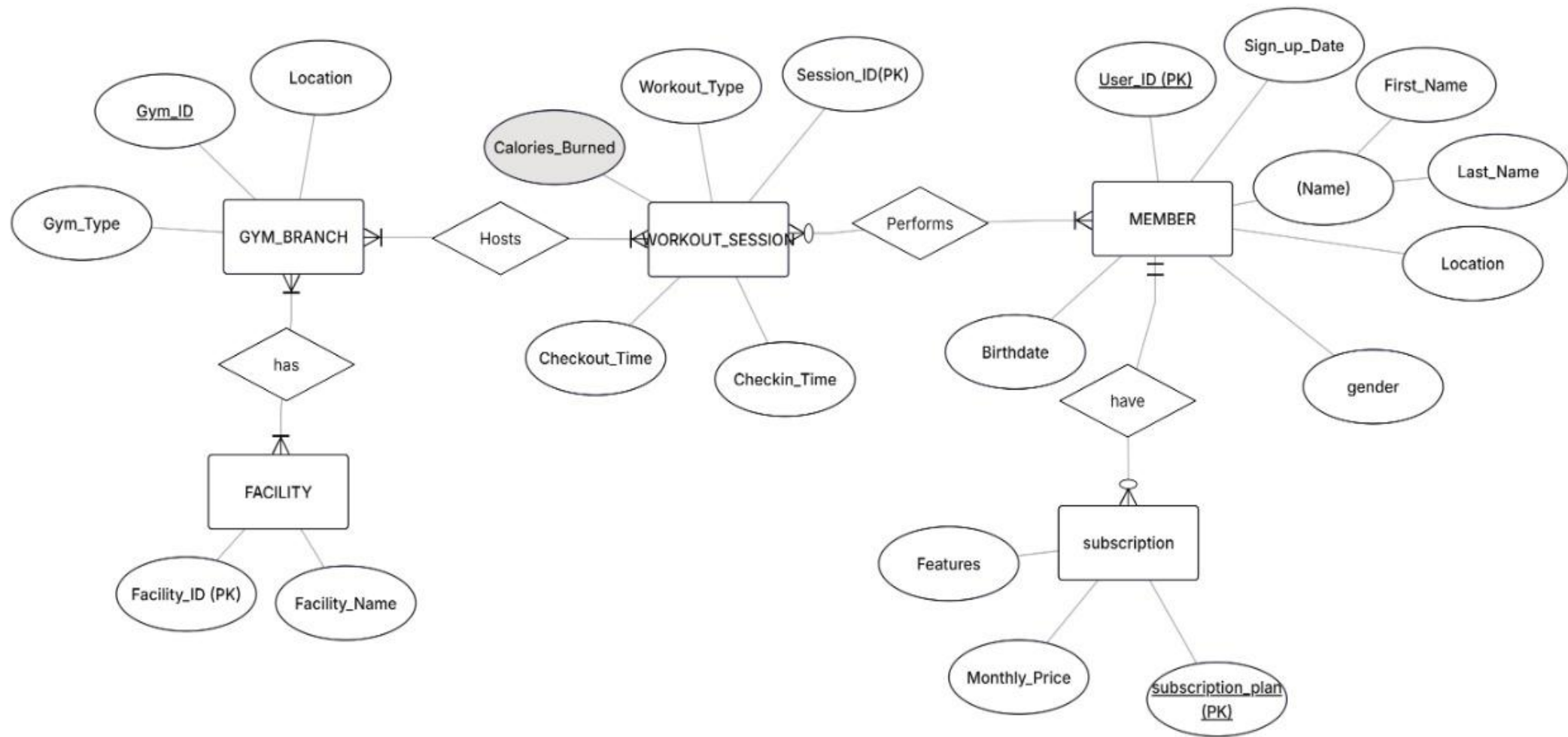
DAX Measures

Visualization

Power BI

Gym Management System & Analytics Project - Scenario Document

database for a Gym Management System is used to manage members, gym branches, subscriptions, and workout sessions. The gym company has many branches; each branch has a unique Gym ID, location, gym type, and available facilities. The system stores information about members including a unique User ID, first name, last name, age, gender, birthdate, sign-up date, location, and subscription plan. Each member subscribes to one subscription plan. Each subscription plan is identified by a unique Subscription Plan and includes the monthly price and available features. A member can visit any gym branch and perform different workout sessions. The system records each workout session including the Gym ID, User ID, check-in time, check-out time, workout type, and calories burned. The system allows the gym management to monitor members' activities, analyze workout behavior, evaluate branch performance, and generate reports for better decision-making.



Data Cleaning

Why?

Before analyzing member behavior and branch performance, the collected gym data must be cleaned and prepared. Raw records often contain missing values, duplicates, inconsistent formats, and logical errors. Data cleaning ensures accuracy and reliability, allowing meaningful insights to be generated and supporting better decision-making.

Injected Data Quality Issues

Simulated on Workout_Session • 300,000 → 304,500 rows (after duplicates) • seed = 42

Issue	Column	% Rows	Cleaning Stage
Missing values (NULL)	calories_burned	3.0%	Python
Missing values (NULL)	checkout_time	2.0%	Excel
Duplicate rows	all columns	1.5%	SQL
Inconsistent casing / whitespace	workout_type	5.0%	SQL
Spelling typos (Yogaa, Crosfit)	workout_type	2.0%	Excel
Inconsistent date format	checkin_time	3.0%	Excel
Outlier / invalid values	calories_burned	1.0%	Python
Logical error (checkout < checkin)	checkout_time	1.0%	SQL
Extra leading/trailing whitespace	user_id	2.0%	SQL

Multi-Stage Cleansing Pipeline

1

Partial SQL Clean

Duplicates, casing/whitespace,
logical checkout errors



2

Excel & Power Query

Missing checkout_time,
spelling typos, date formats



3

Python — 100% Clean

Missing & outlier
calories_burned values

Re-imported into SQL for summarization (joins, aggregations, subqueries) before modeling and visualization.

Data Quality: Before vs After

Before Cleaning

305K

Dirty Row Count

5K

Duplicate Count

9K

Null Calories

6K

Null Checkout

After Cleaning

297K

Clean Row Count

0

Duplicate Count

0

Null Calories

0

Null Checkout

All nine injected data-quality issues were fully resolved across the three cleansing stages, producing a fully reliable dataset for analysis.

Data Model Blueprint



Gym branches host workout sessions and offer facilities; members perform sessions and subscribe to a plan.

Key Insights — Branch

1

Member Distribution

Member counts are relatively balanced across the five gym branches, with no single branch dominating.

2

Peak Hours

Most check-ins occur during daytime hours, while early-morning and late-night activity remains low.

3

Monthly Attendance Trend

Attendance is generally stable month over month; the sharp dip in the final month reflects incomplete data rather than an actual decline.

4

Calories Burned by Gym Type

Budget, Standard, and Premium branches show very similar median calories burned — gym type has little effect on workout intensity.

Key Insights — Member

1

Age Distribution

Members are spread fairly evenly across the 18–64 age range, with no dominant age group.

2

Gender Distribution

Male 46.6%, Female 43.6%, Non-binary 9.8% — a well-balanced member base.

3

Calories by Subscription Plan

Basic, Student, and Pro plans show similar calorie-burn distributions; a higher-tier plan does not necessarily mean higher effort.

4

Monthly Sign-ups

Sign-ups remain stable throughout the period, with the final month's dip attributable to incomplete records.

5

Age vs. Calories Correlation

Pearson correlation = 0.0 — no meaningful linear relationship between age and calories burned.

KPIs & Dashboard Overview

5,000

Total Members

~297K

Total Sessions

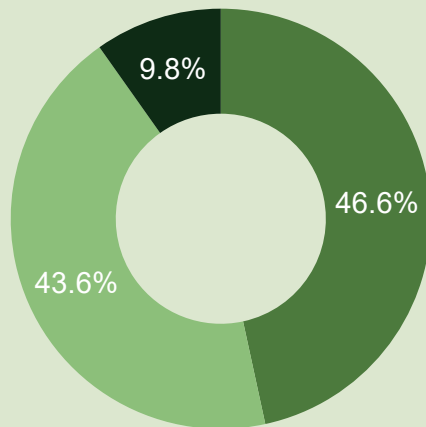
883.77

Avg. Calories Burned

262M

Total Calories Burned

Members by Gender



■ Male ■ Female ■ Non-binary

Dashboard Highlights

- Power BI dashboard with Before/After cleaning quality checks
- Sessions by Branch, Members by Gender, Workout Types by Plan
- Interactive filters for branch, subscription plan, and time period

Data Quality Check - Before Cleaning

305K

Dirty Row Count

5K

Dirty Duplicate Count

9K

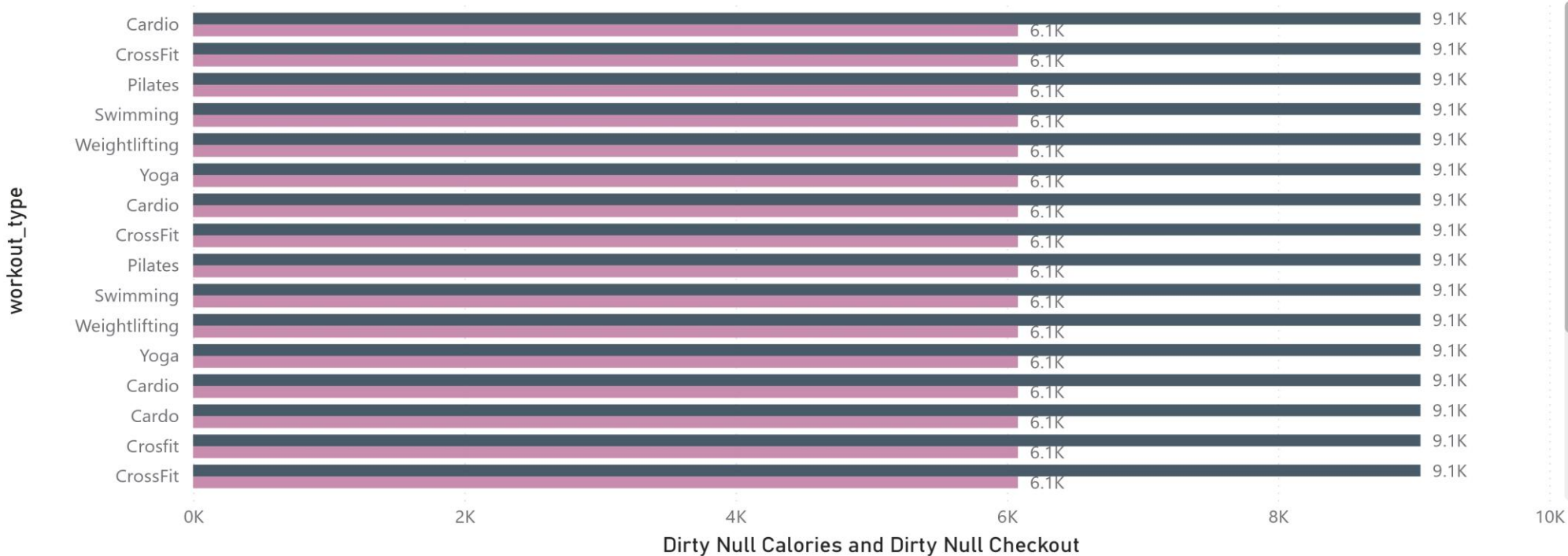
Dirty Null Calories

6K

Dirty Null Checkout

Dirty Null Calories and Dirty Null Checkout by workout_type

● Dirty Null Calories ● Dirty Null Checkout



Data Quality Check - After Cleaning

297K

Clean Row Count

0

Clean Duplicate Count

0

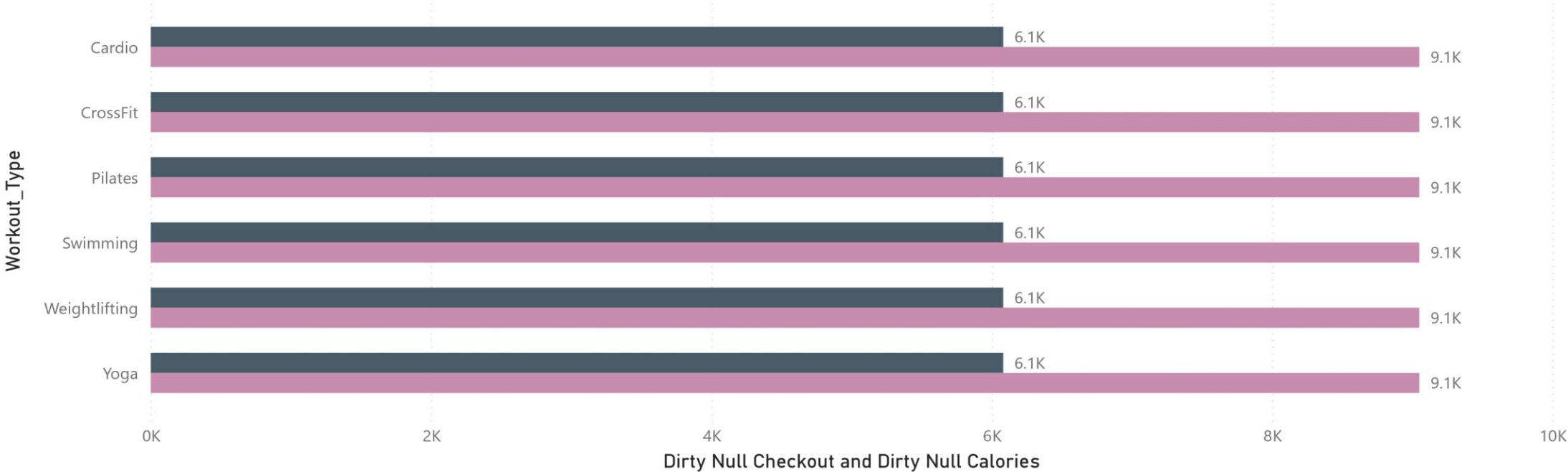
Clean Null Checkout

0

Clean Null Calories

Dirty Null Checkout and Dirty Null Calories by Workout_Type

● Dirty Null Checkout ● Dirty Null Calories



Gym Management System - Analysis Dashboard

5K

Total Members Fixed

297K

Total Sessions Fixed

883.77

Avg Claories Burned

262M

Total Calories Fixed

Branch

Row Labels

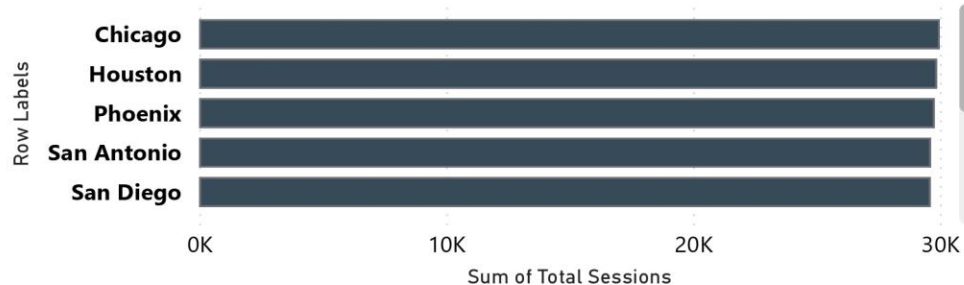
All

Subscription Plan

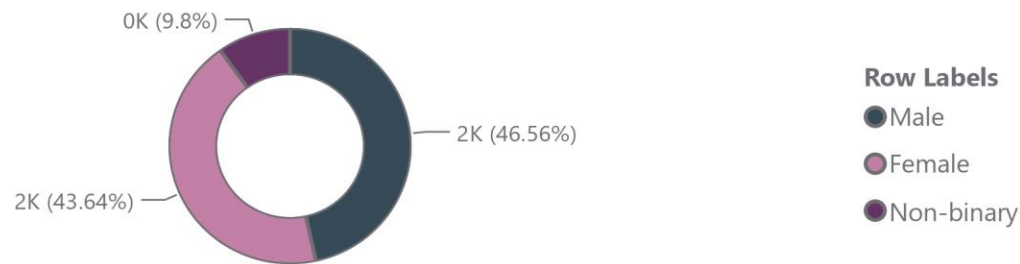
Row Labels

All

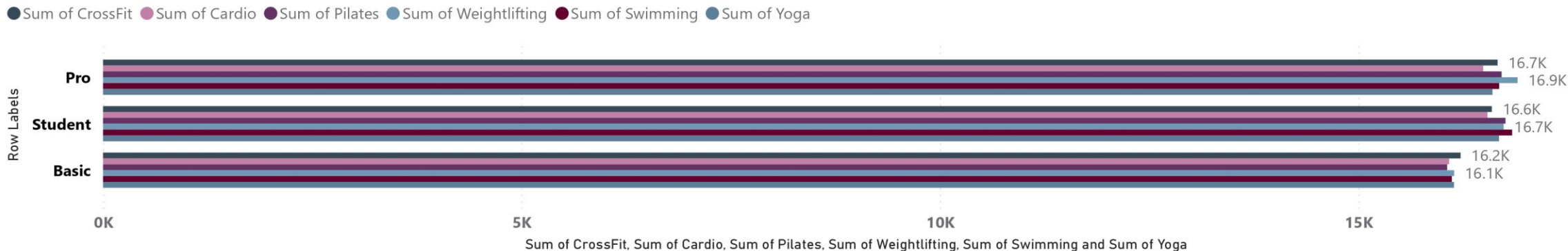
Sessions by Branch



Members by Gender



Workout Types by Subscription Plan



Business Recommendations

1

Focus on improving workout programs and training quality rather than relying on gym type or subscription tier to drive results.

2

Continue marketing to a broad age range, since the customer base is diverse across the 18–64 span.

3

Maintain balanced marketing strategies across all genders, given the well-balanced member base.

4

Collect additional variables — workout duration, exercise frequency, and attendance patterns — to enable deeper analysis.

Conclusion

- A fully reliable dataset of ~297K clean workout sessions across 5,000 members was produced after a rigorous 3-stage cleansing pipeline.
- Member engagement is broad-based: age, gender, and subscription tier show no strong bias toward higher activity or calorie burn.
- Branch performance is balanced, with daytime hours consistently driving the highest attendance across all locations.
- Gym type and subscription tier are not strong predictors of workout intensity — program quality matters more than pricing tier.

Thank You!

We sincerely thank NTI for their continuous guidance and support throughout this project, and we extend our appreciation to our supervisor for the valuable feedback and encouragement provided along the way.